

Modeling corticotroph deficiency with pituitary organoids supports the functional role of *NFKB2* in human pituitary differentiation

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

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Reviewed preprint posted

November 7, 2023 (this version)

Posted to bioRxiv

August 4, 2023

Sent for peer review

July 31, 2023

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Abstract

Background

Deficient Anterior pituitary with common Variable Immune Deficiency (DAVID) syndrome is a rare condition characterized by the association of adrenocorticotrophic hormone deficiency (ACTHD) and primary hypogammaglobulinemia, caused by *NFKB2* heterozygous mutations. Nuclear factor kappa B (NFKB) signaling is a key regulator of the immune system; however, the underlying mechanism of its association with endocrine symptoms remains unknown. Two main hypotheses explain the effects of mutant *NFKB2* on the pituitary gland: an autoimmune hypophysitis, preferentially affecting corticotroph function, or a primary developmental defect. The role of *NFKB2* in the development of the human pituitary was called into question by *Nfkb2*-deficient *Lym1* mice, which have normal pituitary functions.

Purpose

The aim of this study was to create a human disease model to define the role of *NFKB2* in human pituitary development.

Methods

We established pituitary organoids in three dimensions (3D) culture after directed differentiation from CRISPR/Cas9-edited human induced pluripotent stem cells (hiPSC). First, we conducted a proof-of-concept study, introducing a homozygous *TBX19*^{K146R/K146R} missense pathogenic variant in hiPSC, an allele found in patients with congenital isolated

ACTHD. Then, we used the same method to produce *NFKB2*^{D865G/D865G} mutant organoids, harboring the pathogenic missense variant previously identified in DAVID patients. This mutation causes a failure of NFKB2 p100 phosphorylation that blocks processing to form active NFKB2 p52. We then characterized pituitary organoid development by transcriptomics using bulk RNA sequencing and quantitative RT-PCR, and by immunofluorescence in section and whole-mount.

Results

Analysis of wild-type (WT) organoids demonstrated that this *in vitro* model recapitulates corticotroph cell differentiation. *TBX19*^{K146R/K146R} organoids conserved early expression of *HESX1*, but had significantly decreased *PITX1*, *TBX19*, *LHX3*, and *POMC* transcription. *NFKB2*^{D865G/D865G} organoids also had dramatically reduced corticotrophs. Furthermore, *NFKB2*^{D865G/D865G} perturbs the normal expression of 66 genes known to contribute to pituitary development, among which 21 transcription factors.

Conclusions

We used a combination of CRISPR/Cas9 editing and refinement of a 3D organoid culture protocol to model human ACTHD due to *TBX19* or *NFKB2* mutations. The *NFKB2* variant studied induced a significant decrease in corticotroph differentiation, demonstrating for the first time a direct functional role of NFKB2 in human pituitary development. Signaling through NFKB2 is thus a valid new candidate pathway in the pathogenesis of isolated or syndromic ACTHD.

eLife assessment

This **valuable** study examines the effects of NFKB2 mutations on pituitary gland development through hypothalamic-pituitary organoids. The evidence supporting the main conclusions is **solid**, although analysis of additional clones to exclude inter-clone variability would strengthen the conclusions. Insight into the mechanism of action of NFKB2 during pituitary development is **incomplete**. This work will be of interest to endocrinologists and biologists working on pituitary gland development and disease.

Introduction

Adrenocorticotrophic hormone deficiency (ACTHD) is defined by an insufficient production of ACTH by the pituitary, followed by low adrenal cortisol production. Proper diagnosis and management of ACTHD is crucial, as it is a life-threatening condition in the neonatal period, characterized by hypoglycemia, cholestatic jaundice and seizures.¹ Constitutional ACTHD, which is diagnosed at birth or during the first years of life, can be isolated or associated with deficiencies in other pituitary hormones such as growth hormone (GH) or thyrotropin-stimulating hormone (TSH) and termed combined pituitary hormone deficiency (CPHD). ACTHD is genetically heterogeneous, due to mutations in genes coding mostly for transcription factors responsible for

pituitary ontogenesis or their regulation. Responsible genes identified to date include the T-box transcription factor *TBX19* (also known as *TPIT*), *NFKB2*, *LHX3*, *LHX4*, *PROP1*, *HESX1*, *SOX2*, *SOX3*, *OTX2* and *FGF8*.

Mutations of *TBX19* account for approximately two-thirds of neonatal-onset complete isolated ACTHD¹. *TBX19* is a T-box transcription factor restricted to pituitary pro-opiomelanocortin (POMC)-expressing cells in mice and humans. It is essential for *POMC* gene transcription and terminal differentiation of POMC-expressing cells in the anterior pituitary (also known as the adenohypophysis)². POMC is cleaved to produce ACTH in corticotroph cells. *TBX19*-deficient mice have only a few pituitary *POMC*-expressing cells, with very low ACTH and undetectable corticosterone levels³. In humans with isolated ACTH deficiency, *TBX19* mutations lead to loss-of-function by different mechanisms, such as the K146R (exon 2, c.437 A>G) pathogenic variant located in the T-box region, which results in a loss of DNA-binding ability¹.

Thanks to GENHYPOPIT^{4–6}, an international network aimed at identifying new genetic etiologies of combined pituitary hormone deficiency, we described the first cases of DAVID (Deficient Anterior pituitary with common Variable Immune Deficiency) syndrome, a rare association of hypopituitarism (mainly ACTHD) and immune deficiency (hypogammaglobulinemia)⁷. DAVID syndrome is associated with variants in the nuclear factor kappa-B subunit 2 (*NFKB2*) gene^{8,9}. In particular, we reported that a pathogenic, heterozygous D865G (exon 23, c.2594 A>G) *NFKB2* variant was found in a patient presenting with severe recurrent infections from 2 years of age, who at the age of 5 was diagnosed with ACTHD⁷. Mutations in the C-terminal region of *NFKB2* lead to the disruption of both non-canonical and canonical pathways^{10–12}.

Full-length *NFKB2* (p100) protein has 2 critical serines, S866 and S870, in the C-terminal domain. Phosphorylation of these sites yields the transcriptionally active form of *NFKB2* (p52) by enabling the proteasomal processing of the larger p100 protein. The D865G mutation, located adjacent to the critical S866 phosphorylation site, protects mutant protein from proteasomal degradation, causes defective processing of p100 to p52, and results in reduced translocation of p52 to the nucleus^{8,11,12}. *NFKB2*^{D865G/+} resulted in half of the normal level of processing to p52, whereas *NFKB2*^{D865G/D865G} exhibited near-absence of p52¹². The nuclear factor kappa B (*NFKB*) signaling pathway is a key regulator of the immune system^{12–14}, which likely explains the immune phenotype of patients with DAVID syndrome, including susceptibility to infections and auto-immune disorders¹⁵. In contrast, the underlying mechanism causing pituitary disorders remains unknown, with two predominating hypotheses: an indirect autoimmune hypophysitis, preferentially affecting corticotroph function, or a primary developmental defect supported by the expression of *NFKB2* in the human fetal pituitary¹⁶. However, a developmental role for *NFKB2* was not confirmed in the *Lym1* mouse model, carrying a heterozygous or homozygous nonsense variant Y868*, which presented apparently normal pituitary development and function although impairing immunity⁹. Although the mouse continues to be a relevant model for many aspects of pituitary organ development, we sought a higher-throughput method to assess pathogenic effects of new candidate genes or variants on human pituitary differentiation.

Human induced pluripotent stem cell (hiPSC)-derived organoids have emerged as promising models to study many developmental mechanisms and their perturbation in disease¹⁷. Pioneering work on human embryonic stem cells and later, hiPSC, established that 3D organoid models can replicate aspects of pituitary development^{18–20}, and could thus be of major interest in modeling pituitary disorders^{18,21,22}. In the present study, we applied recent progress in genome editing using CRISPR/Cas9²³ to a refined protocol to derive 3D organoids from hiPSC in order to model ACTHD. We validated the recapitulation of corticotroph cell differentiation *in vitro* by introducing a *TBX19* mutation known to induce isolated human ACTHD and documenting the subsequent deficiencies in corticotroph development. When used to characterize the D865G *NFKB2* variant found in patients with DAVID syndrome, hiPSC-derived

pituitary organoids displayed dramatically altered corticotroph differentiation in the absence of immune cells, demonstrating for the first time a direct and incontrovertible role for NFKB2 in human pituitary development.

Materials and methods

Culture and maintenance of hiPSC lines

The 10742L hiPSC line, derived from a healthy individual, was used in the study as the WT control (characterized and provided by the Cell Reprogramming and Differentiation Facility [MaSC], Marseille Medical Genetics, Marseille, France). Further information about this line is indicated in **Suppl Table S1**. Two mutant lines carrying *TBX19*^{K146R/K146R} and *NFKB2*^{D865G/D865G} were subsequently generated from 10742L using CRISPR/Cas9 editing as described below. All hiPSC lines were cultured on six-well plates (Corning, #3335, New York, USA) coated with Synthmax II-SC Substrate (working concentration at 0.025 mg/mL, Corning, New York, USA) and maintained undifferentiated in a chemically defined growth medium (StemMACs hiPSC-Brew XF ; MACS Miltenyi Biotec, Paris, France)²⁴. hiPSC lines were maintained in a humidified incubator under conditions of 37°C, 5% CO₂, with a daily change of medium, and passaged when cells reached 60–80 % confluency using enzyme-free ReleSR, according to manufacturer recommendations (StemCell Technologies, Canada, #05872).

CRISPR/Cas9 mediated genome editing of hiPSC

CRISPR/Cas9 gene editing was used to introduce the *TBX19*^{K146R/K146R} and the *NFKB2*^{D865G/D865G} mutations using a previously described method²⁵. The *TBX19*^{K146R/K146R} line harbored the c.437A>G, p.Lys146Arg *TBX19* (NM_005149) pathogenic missense variant. The *NFKB2*^{D865G/D865G} line harbored the c.2594A>G, p.Asp865Gly *NFKB2* (NM_001322934) pathogenic missense variant.

Preparation of the CRISPR/Cas9 sgRNA plasmid

For each targeted gene, a sgRNA plasmid was prepared as previously described²⁶. Briefly, we designed a sgRNA sequence within 20 nucleotides of the target site, selected with the open-source CRISPOR tool (<http://crispor.tefor.net/>)²⁷.

The sgRNA sequence used to generate *TBX19*^{K146R} was (5'>3'): AAGCTGACCAACAAGCTCAA (see also Table S2).

The sgRNA sequence used to generate *NFKB2*^{D865G} was (5'>3'): GTGAAGGAAGACAGTGCCTA (see also Table S3).

We used the cAB03 (also known as pX459-pEF1alpha) vector as previously described²⁵. In this, the pSpCas9 (BB)-2A-Puro (PX459) V2.0 (Addgene plasmid # 62988, Addgene, Teddington, UK) was modified by an EF1alpha promoter²⁵. The chosen sgRNA was cloned into the cAB03 open plasmid vector as previously described by Arnaud et al²⁵. The final vector expressed the corresponding sgRNA under the control of the U6 promoter, as well as Cas9 under the control of the EF-1alpha promoter, and a puromycin resistance gene.

Single-stranded oligo DNA nucleotides (ssODN) design

We also designed donor sequences to generate *TBX19*^{K146R} and *NFKB2*^{D865G} missense mutations. A donor sequence is used for homology-directed repair, allowing the introduction (“knock-in”, KI) of single nucleotide variations with low efficiency. To optimize the efficiency of KI editing, the donor sequences were designed as single-stranded oligo DNA nucleotides (ssODN), carrying the mutation of interest, and also included a silent (or blocking) mutation that mutates the

protospacer-adjacent motif (PAM) or disrupts the sgRNA binding region to prevent re-cutting by Cas9 after successful editing^{25, 28–30}. The ssODN contained 100 nucleotides with homology arms on each side of the target region. The ssODN sequences were as follows (see also **Suppl Tables S2 – Table S3**):

For *TBX19*^{K146R} (5'-3'):

TGGATGAAAGCTCCCATCTCCTTCAGCAAAGTGAGGCTGACCAACAAGTTAAATGGAGGCGG
GCAGGTACGAATGAGGCGGGCAGGCCTGGCCACCCGCT

For *NFKB2*^{D865G} (5'-3'):

TCCCATTCTGTCCCCATTACCCCCAGCAGAGGTGAAGGAAGGCAGTGCCTACGGGAGCCAG
TCAGTGGAGCAGGAGGCAGAGAAGCTGGGCCACCCC.

ssODN were synthesized by Integrated DNA Technologies (Coralville, Iowa, US) at Ultramer™ quality.

Electroporation

hiPSC were transfected with constructed sgRNA/Cas9 vectors by electroporation using the Neon Transfection System 100 µL kit (Invitrogen). A summary of the procedure is described in **Suppl Figure S1**.

Clone isolation and screening

On day 3 post-transfection, we used the cleaved amplified polymorphic sequences (CAPS) assay to estimate the efficacy of CRISPR/Cas9 in bulk transfected hiPSC populations and to choose a number of colonies to be picked for further amplification³¹. A restriction enzyme recognition site was included in each donor template (BseI for *TBX19* and MseI for *NFKB2*). CAPS primers and restriction enzymes were chosen assisted by AmpliX software (version 2.0.0b3; <https://inp.univ-amu.fr/en/amplifx-manage-test-and-design-your-primers-for-pcr>). CAPS primers are listed in **Suppl Table S4**. The PCR reaction was performed using the PCR condition reported in **Suppl Table S5 – Suppl Table S6**. The CAPS products were separated and visualized by electrophoresis on a 2% agarose gel and analyzed for densitometry with ImageJ^{25, 30}.

After 10 days, the CAPS assay was also used to screen and identify possible knock-in (KI) clones. KI clones were then confirmed by Sanger sequencing (Genewiz, Leipzig, Germany). Analyses of Sanger traces were made using Sequencher DNA sequence analysis software (version 5.4.6, Gene Codes Corporation, Ann Arbor, MI USA, <http://www.genecodes.com>) to align sequences of KI clones with WT sequence to confirm the successful edition. Sanger sequencing primers are described in **Suppl Table S4**. KI clones were amplified and then cryopreserved.

In vitro pituitary organoids differentiated from hiPSC

Pituitary organoids were produced in parallel using the same protocol for WT and edited hiPSC lines. Two batches of differentiation with two hundred organoids for each hiPSC line were generated, respectively WT versus *TBX19*^{K146R/K146R} organoids or WT versus *NFKB2*^{D865G/D865G} organoids.

Differentiation was based on a protocol from Matsumoto et al. with some modifications^{20, 22}. hiPSCs were first dissociated into single cells using Accutase. Ten thousand cells per organoid were plated in 20 µL hanging drops under the lid of a 127.8 x 85.5mm single-well plate (Greiner bio-one) in growth factor-free, chemically defined medium (gFCDM) supplemented with 10% (vol/vol) Knockout Serum Replacement (KSR; Thermo Fisher Scientific, #10828-010) and 20 µM Y-27632 (Sigma-Aldrich, # SCM75). The lid was inverted over the bottom chamber (filled with 10 mL of sterile water to avoid the evaporation of droplets) and incubated at 37°C/5% CO₂ for 2 days. Once

aggregates formed, organoids were transferred to non-adhesive 100x20mm culture dishes (Corning) containing 10 mL of gfCDM medium supplemented with 10% KSR without Y-27632 and incubated at 37°C and 5% CO₂ until day 6. From days 6 to days 17, the medium was supplemented with 10% KSR, 5 nM recombinant human bone morphogenetic protein 4 (BMP4; Peprotech, # AF-120-05ET, USA) and 2 µM Smoothed agonist (SAG; Sigma-Aldrich/Merck, # SML 1314). Half of the medium was renewed every 2 - 3 days.

From day 18, BMP4 was withdrawn and half of the medium was renewed every 2 - 3 days. At this point, organoids were maintained under high-O₂ conditions (40%) and 5% CO₂ in an MCO-5M incubator (Panasonic). From day 30, gfCDM medium supplemented with 20% knockout serum replacement was used, and all medium was renewed every 2-3 days until at least day 105. Induction into pituitary progenitor cells (defined as LHX3+) and into pituitary hormone-producing cells was evaluated by qRT-PCR (on days 0, 6, 18, 27, 48, 75, 105) and immunofluorescence (on days 48 and 105) (see below).

Validating *TBX19*^{K146R} targeting and assessing off-target mutations by whole genome sequencing (WGS)

For *TBX19*^{K146R}, we used whole genome sequencing (WGS) to assess on-target and off-target mutations, because the edited site is far from the cut site (15 bases). DNA extracted from control and *TBX19*^{K146R/K146R} (5 organoids each) was submitted for WGS (Integrigen Genomics platform). PCR-free libraries were prepared with the NEBNext UltraTM II DNA Library Prep Kits (New England BioLabs) according to supplier recommendations. Specific double-strand gDNA quantification and a fragmentation (300 ng of input with high-molecular-weight gDNA) sonication method were used to obtain approximately 400 bp fragments. Finally, paired-end adaptor oligonucleotides (xGen TS-LT Adapter Duplexes from Integrated DNA Technologies) were ligated and re-paired. Tailed fragments were purified for direct sequencing without a PCR step. DNA PCR free libraries were sequenced on paired-end 150 pb runs on the NovaSeq6000 (Illumina) apparatus. Image analysis and base calling were performed using Illumina Real Time Analysis (RTA) Pipeline with default parameters. Sequence reads were mapped on the Human Genome Build (GRCh38) using the Burrows-Wheeler Aligner (BWA)³². Variant calling was performed via the GATK Haplotype Caller GVCF tool (GATK 3.8.1, Broad Institute)³³. Mutation enrichment was determined using Fisher's Exact Test. Variants were annotated with Ensembl's Variant Effect Predictor, VEP) by Integrigen Genomics³⁴. These NGS analyses confirmed the presence of on-target mutation and the absence of off-target mutation in the top four predicted off-target sites.

Validating *NFKB2*^{D865G} targeting, assessing off-target mutations and differential expression analysis by bulk RNA sequencing (RNA-seq)

For *NFKB2*^{D865G}, we used bulk RNA sequencing (RNA-seq) for analysis of differential RNA expression, and to assess on-target and off-target mutations, because the target site is close to the cleavage site (6 bases). A total of 500 ng of total RNA was extracted from 10 individual organoids (5 *NFKB2*^{D865G/D865G} and 5 control organoids). RNA-seq libraries were generated using the KAPA mRNA HyperPrep kit (Roche). The quality and profile of the libraries were visualized on a Bioanalyzer using the High Sensitivity DNA assay (Agilent) and quantified on a Qubit using the dsDNA High Sensibility Kit (Thermo Fisher Scientific). Finally, we performed 2x 76-bp paired-end sequencing on a NextSeq500 (Illumina). After quality control using FastQC (Braham Institute), reads were aligned on the GRCh38 human reference genome using STAR (version 2.7.2b)³⁵. BAM files were ordered and indexed using Samtools³⁶. Read counts on genes were determined using StringTie³⁷. Genes differentially expressed between conditions were identified using the R package DESeq2 (version 1.34.0)³⁸. Genes with an adjusted P-value < 0.05 found by DESeq2 were

assigned as differentially expressed. We assessed the presence of desired *NFKB2* edits according to the CRISPR sgRNA design by direct visualization of the BAM files and the absence of off-target coding variations by the GATK Haplotype Caller GVCF tool³³.

Total RNA isolation and quantitative real-time PCR (qRT-PCR) analysis for assessment of mRNA expression of pituitary organoids development

Organoids were harvested, placed on ice and stored dry at -80°C until RNA extraction. The total RNA was extracted from WT, *TBX19*^{K146R/K146R} and *NFKB2*^{D865G/D865G} organoids using the NucleoSpin RNA Plus XS kit (Macherey-Nagel), and measured on a NanoDrop TM 1000 Spectrophotometer (Thermo Fisher Scientific). Organoid RNAs were extracted on days 0, 6 and 18 from 7 or 8 organoids, or on days 27, 48, 75 and 105 from single organoids (3-8 organoids per group for each time point). cDNA was synthesized from 200 ng total RNA using a mix of M-MLV reverse transcriptase (Invitrogen, UK, #28025013), dNTP 10mM (Invitrogen, #10297018), RNaseOut Inhibitor (Invitrogen, #10777019) and random primers (Invitrogen, UK, #48190011). Quantitative PCR was performed using iTaq Universal SYBR Green Supermix (Bio-Rad Laboratories), on a QuantStudio™ 5 Real-time PCR system (Thermo Fisher Scientific) and analysed using QuantStudio™ 5 Design & Analysis Software. The thermal cycling profiles were as follows: initial denaturation at 95°C for 20 seconds, followed by 40 cycles of denaturation at 95°C (3 sec), annealing at 60°C (45 sec) and extension at 72°C (45 sec). All samples were assayed in duplicate. The beta-actin (*ACTB*), glyceraldehyde-3-phosphate dehydrogenase (*GAPDH*) and beta-tubulin (*TUBB*) transcripts expressions were used as three endogenous reference controls. Target gene expressions were normalized relative to the mean of the housekeeping genes using the delta Ct method, and then normalized relative to the highest mean values of WT during the 105 days using the comparative 2^{-ΔΔCt} method³⁹. Primer sequences used in the experiments are shown in **Suppl Table S7**.

Immunofluorescence labeling experiments

At days 48 and 105, ten each of WT and mutant organoids were fixed using 4% buffered paraformaldehyde in standard phosphate-buffered saline (PBS) for an hour at RT before rinsing in PBS. After overnight incubation in 30% sucrose/PBS (w/v) at 4°C, organoids were embedded in Surgipath medium (Leica, # 3801480) and snap frozen. Organoids were cryosectioned at 16 μm. After blocking with 0.01% Triton X100 (Sigma-Aldrich, #329830772), 10% normal donkey serum (Millipore, USA, #S30-100 ML) and 1% fish skin gelatin (Sigma-Aldrich, Canada, #G7765-250 ML) in PBS with 5 ng/mL 4',6-diamidino-2-phénylindole (DAPI) for an hour at RT, slides were incubated with primary antibody diluted in antibody solution (PBS, Triton as above, 1% normal donkey serum and 0.1% fish skin gelatin) overnight at 4°C, washed and incubated and a 1/500 dilution of secondary antibody in PBS for 2 hours at RT. The primary antibodies and dilutions used in this study are summarized in **Suppl Table S8**. Secondary antibodies were species-specific conjugates to Alexa Fluor 488, 555 or 647 (Thermo Fisher Scientific). Fluorescence images were obtained using LSM 800 confocal microscopy (Zeiss).

3D imaging of pituitary organoids

We used a whole-mount immunostaining and clearing protocol adapted from Belle et al⁴⁰. 4-8 organoids of each genotype (WT, *TBX19*^{K146R/K146R} or *NFKB2*^{D865G/D865G}) were fixed by immersion in 4 % buffered paraformaldehyde (PFA) in PBS overnight at 4°C. All steps were carried out on a slowly rotating (70 rpm) agitator. After washing in PBS, organoids were either stored at 4°C or blocked and permeabilized with 5 mL PBSGT for 2 days at RT (PBSGT: 0.2% fish skin gelatin, 0.5% Triton X100 in PBS, NaN₃ 0.01%). Organoids were incubated with primary antibodies in 3 mL PBSGT (at concentrations used on sections) over 5 days at 37°C to increase penetration. Organoids

were then washed 6 times in an excess of PBSGT at RT, before incubation with secondary antibodies in 3 ml PBSGT at RT overnight. The wash step was repeated. Organoids were then embedded in 1% low-melting temperature agarose in 1X TAE after it had cooled to about 45°C.

As adapted from the iDISCO+ protocol⁴¹, organoids in agarose blocks were progressively dehydrated to 100% methanol (VWR, #20847.360) with 1h at each graded step. After overnight incubation in two parts dichloromethane (DCM, Sigma-Aldrich, #270997) to one part 100% methanol, organoids were immersed in 100% DCM for 30 minutes before changing to 100% dibenzyl ether (DBE; Sigma-Aldrich, #108014) for transparization over two hours. Samples were maintained in DBE at RT and protected from light in an amber glass vial. Just before imaging, they were transferred to room-temperature ethyl cinnamate (Sigma-Aldrich, #112372). Light-sheet fluorescence microscopy (Miltenyi Biotec UltraMicroscope Blaze™) was used to acquire z-stacks of optical sections of pituitary organoids at 4 µm intervals. After 3D reconstruction with Imaris software (version 9.6, Bitplane), the number of corticotrophs in each organoid was defined by counting individualized cell-sized (7 µm) objects positive for ACTH immunoreactivity with the Spots automatic classifier after setting parameters in a region of interest containing positively labeled WT cells, then applied to all organoids. Using the Surfaces tool of the Imaris software, the volume of organoids was determined separately, based on thresholds corresponding to their respective fluorescence intensities.

Statistical analysis

Statistical analyses were performed and visualized with GraphPad Prism 9.5.0 software (GraphPad Software, Inc). Data are expressed as means ± SEM. Comparisons between the two groups (mutant versus WT) were performed by unpaired two-tailed t-tests (non-parametric Mann-Whitney test). n refers to the number of samples for each experiment outlined in the figure legends. p values of < 0.05 (*), < 0.01 (**), and < 0.001 (***) were considered statistically significant differences.

Results

Corticotroph deficiency can be modeled by directed differentiation of *TBX19*-mutant hiPSC in 3D culture

To first validate the pituitary organoid model and determine whether the *TBX19* mutant can affect corticotroph differentiation, we generated a homozygous *TBX19*^{K146R/K146R} hiPSC line from the isogenic control line using CRISPR/Cas9 (Figures 1A–1D; Suppl. Figure S1). One KI clone carrying *TBX19*^{K146R/K146R} was obtained after screening 100 clones by CAPS and confirmed by Sanger sequencing (Suppl Figure S2). This mutant clone was then amplified and differentiated into pituitary organoids, in parallel with the control line.

The ability of the *TBX19*^{K146R/K146R} line to differentiate into a hypothalamic-pituitary structure was compared to its control line (200 organoids for each line) using a 3D culture method, in which pituitary-like and hypothalamus-like tissues simultaneously developed (Figures 1E, 2A, Suppl Figure S3A–B). Inside of such organoids, hiPSC differentiate into hypothalamic progenitors, while the outer layer differentiates into an epithelium sharing properties with the oral ectoderm that develops into Rathke's pouch and then the anterior pituitary (Figure 2A, Suppl Figure S3). The expression of several key markers of pituitary development and differentiation was followed in mutant *TBX19*^{K146R/K146R} organoids matched to control WT organoids over time, using qRT-PCR (d0, d6, d18, d27, d48, d75, d105) and immunofluorescence (d48 and d105) (Figure 2A). Organoids grew from 0.4 mm on day 6 to 1.9 mm in their largest 2D dimension by day 105 (Figure 2B).

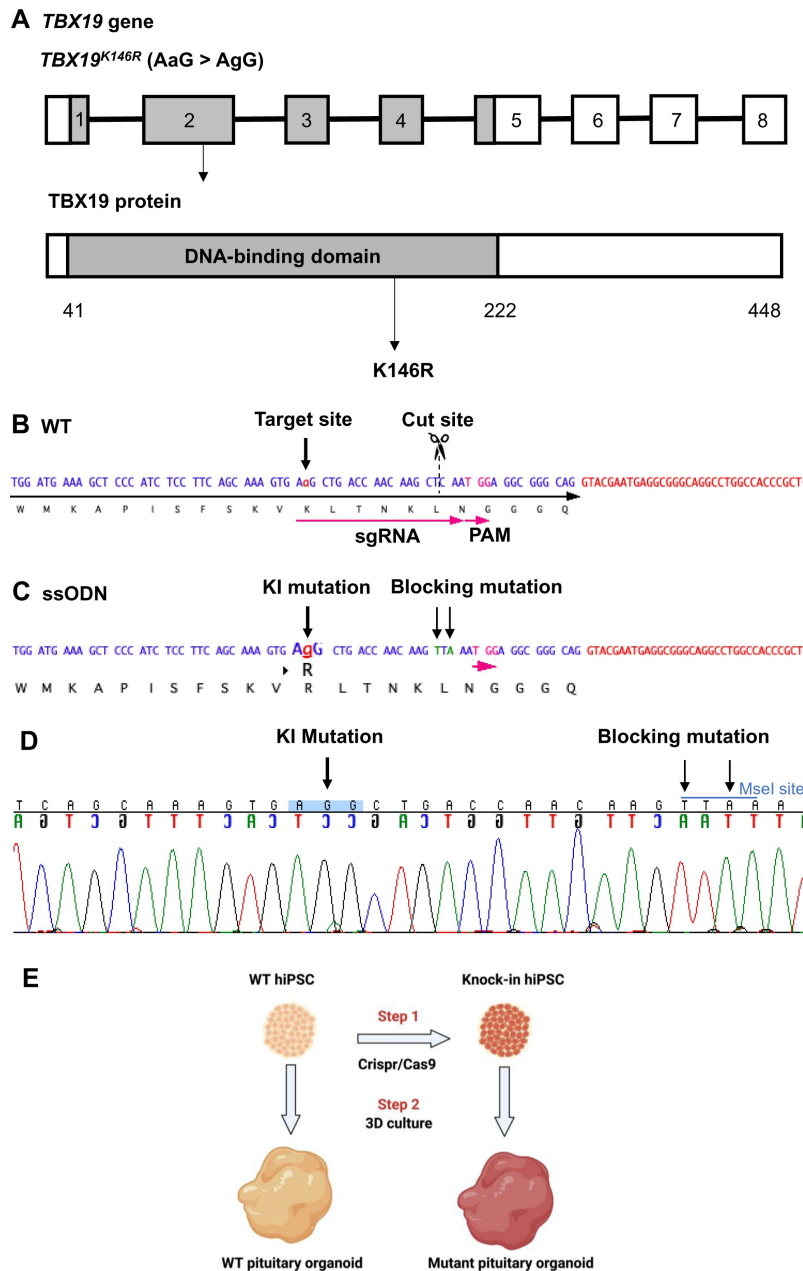


Figure 1

Design of sgRNA and ssODN to edit *TBX19*^{K146R} mutation

A, Illustration of the *TBX19* gene (HGNC: 11596; NCBI: NG_008244.1; Ensembl: ENSG00000143178; Human GRCh38) and *TBX19* protein.

B, Wild-type (WT) sequence containing target site, cut site, target and PAM sequences.

C, ssODN design to edit in a missense K146R mutant of *TBX19* using CRISPR/Cas9.

D, Sequence analysis of a *TBX19*^{K146R/K146R} hiPSC clone 63 obtained by Sanger sequencing after screening by CAPS. This clone was subsequently used in this work to differentiate into pituitary organoids (see below).

E, Summary of our strategy procedure:

Step 1: Production of the knock-in hiPSC lines by CRISPR/Cas9 genome editing.

Step 2: Differentiation into the pituitary organoids from mutant hiPSC lines in parallel with WT line using 3D culture, then comparison of the development of organoids between the two groups.

CAPS: cleaved amplified polymorphic sequences; hiPSC: human induced pluripotent stem cells; KI: knock-in; PAM: protospacer-adjacent motif; sgRNA: single guide RNA; ssODN: single-stranded oligo DNA nucleotides; WT: wildtype; 3D: three dimensions.

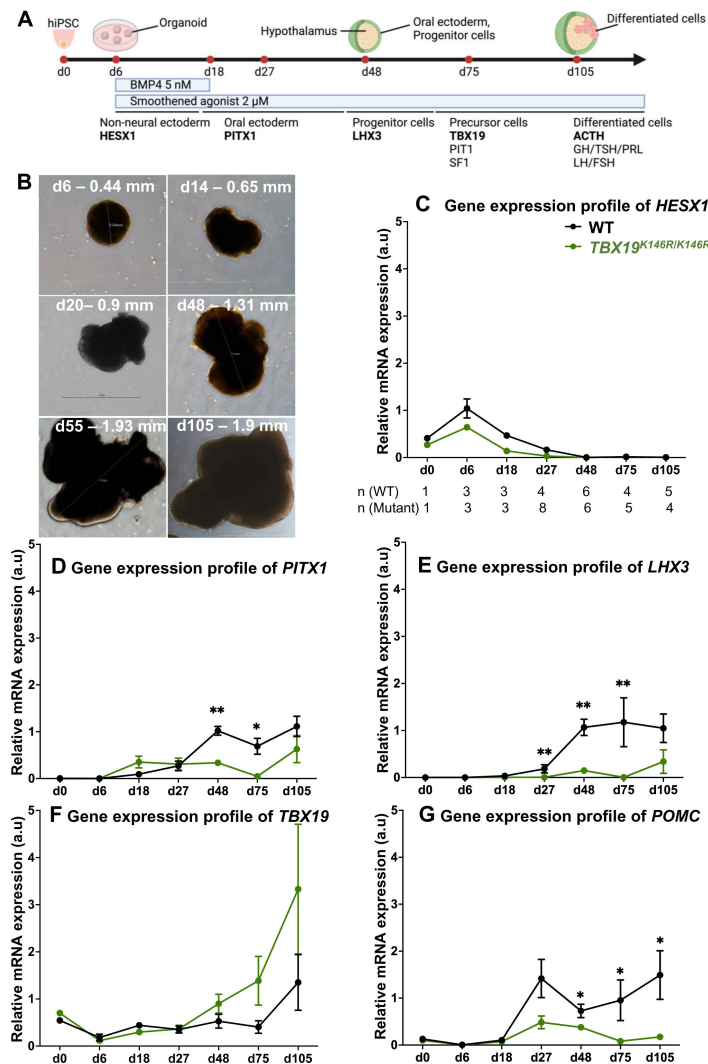


Figure 2

Time course of organoid growth and gene expression in WT and *TBX19*^{K146R/K146R} organoids

A, Culture protocol and schematic to generate pituitary organoids in three-dimensional (3D) culture from hiPSC. Organoids were collected at days (d) 0, 6, 18, 27, 48, 75, and 105 during differentiation to analyze.

B, Bright-field microscopy views of WT organoid examples at different time points throughout differentiation. Scale bars are indicated in each image.

C-G, Relative quantification (RQ) mRNA expression analysis for key markers of pituitary organoids during differentiation: WT (in black line) and *TBX19*^{K146R/K146R} organoids (in green line). Data show means $\pm$ standard error of the mean (SEM); n=1 on day (d)0, n=3 samples per group (8 organoids per sample) at the time points d6 and d18, n=4-8 organoids per group at each subsequent time point (d27, d48, d75 and d105). Mann-Whitney t-test (unpaired, two-tailed, nonparametric). $p < 0.05$ (*), $p < 0.01$ (**).

C, Relative quantification expression of *HESX1*, one of the earliest pituitary placode markers. The expression of *HESX1* was not significantly different in *TBX19*^{K146R/K146R} organoids vs. WT.

D, Relative quantification expression of *PITX1*, an anterior pituitary progenitor marker in the ectodermal primordium. *PITX1* was significantly downregulated in *TBX19*^{K146R/K146R} organoids by d48 (** $p = 0.0022$) and d75 (* $p = 0.0159$).

E, Relative quantification expression of *LHX3*, a pituitary progenitor transcription factor. *LHX3* was significantly lower in *TBX19*^{K146R/K146R} organoids as compared to WT on d27 (** $p = 0.004$), d48 (** $p = 0.0022$) and d75 (** $p = 0.0079$).

F, Relative quantification expression of *TBX19*, a corticotroph lineage marker. There was no significant difference in *TBX19* expression between the two groups ($p = 0.195$, d75).

G, Relative quantification expression of *POMC*, a differentiated corticotroph marker. *POMC* was significantly downregulated in *TBX19*^{K146R/K146R} organoids by d48 (* $p = 0.026$) and thereafter (* $p = 0.0159$ on d75; * $p = 0.0317$ on d105).

Successful differentiation was validated in WT organoids by qRT-PCR, at early stages with the expression of a set of pituitary transcription factors, including *HESX1*, *PITX1* and *LHX3*, and at the latest stage by the expression of corticotroph cell markers *TBX19* and *POMC* (**Figures 2A** [↗](#), **2C-2G**). *HESX1* is the first specific transcription factor of the prospective pituitary and its expression is vital for the early determination of the gland. In WT organoids, *HESX1* was expressed on day 6 and rapidly downregulated as of day 18 (**Figure 2C** [↗](#)). Expression of *PITX1* (a Rathke's pouch marker) and *LHX3* (a pituitary progenitor marker) was high as of day 27, then reached stable levels from d48 onwards (**Figures 2D** [↗](#)-**2E** [↗](#)). Importantly, immunofluorescence demonstrated effective generation of pituitary progenitors in WT organoids with the appearance of *LHX3*⁺ cells in the oral ectoderm-like epithelium by day 48 (**Figure 3A** [↗](#), **Suppl Figure S3A**). This epithelium also expressed *PITX1* and E-cadherin (**Suppl Figure S3C**). Hypothalamus progenitors expressed *NKX2.1* (**Suppl Figure S3B**). By day 105, we observed that WT organoids contained many differentiated corticotroph cells, co-expressing *TBX19* protein in the nucleus and ACTH in the cytoplasm as seen in confocal microscopy, a critical feature for the rest of our investigations (n=8, **Figure 3B** [↗](#), **Suppl Figure S3D**). Consistent with these images, qRT-PCR results confirmed the highest levels of *TBX19* and *POMC* transcription at day 105 in WT organoids (**Figures 2F** [↗](#)-**2G** [↗](#)). Taken together, these data showed that pituitary organoids in 3D culture mimic important aspects of human pituitary ontogenesis, and were characterized by the ability to differentiate into pituitary progenitors by day 48 and then to corticotroph cells by day 105.

We then tested whether *TBX19*-mutant organoids could be used to model ACTHD. To this end, pituitary organoids carrying *TBX19*^{K146R/K146R} were cultured in parallel with WT organoids and analyzed using qRT-PCR and immunofluorescence for the pituitary ontogenesis markers as described above. Our data showed that there was no significant difference in the expression of *HESX1* (**Figure 2C** [↗](#)). However, *PITX1* expression was significantly decreased by days 48 (p= 0.0022) and 75 (p= 0.0159) in mutant organoids (**Figure 2D** [↗](#)). There were significantly lower levels of *LHX3* expression (p= 0.004 on day 27, p= 0.0022 on day 48 and p= 0.0079 on day 75 (**Figure 2E** [↗](#)), although there was no significant change in the expression of the *TBX19* transcript itself (p= 0.195 on day 75, **Figure 2F** [↗](#)). Importantly, *POMC* expression was significantly decreased in the *TBX19*^{K146R/K146R} organoids by days 48 (p= 0.026), 75 (p= 0.0159) and 105 (p= 0.0317) (**Figure 2G** [↗](#)).

Next, we checked several pituitary markers by immunofluorescence (**Figures 3A** [↗](#)-**3D** [↗](#)). In line with the qRT-PCR results, *TBX19*^{K146R/K146R} organoids immunostaining on day 48 showed that *LHX3* protein expression was decreased, in contrast to WT organoids (n= 10 organoids for each group, **Figures 3A** [↗](#), **3C**). By day 105, we observed that there were fewer corticotroph cells expressing ACTH and *TBX19* protein in *TBX19*^{K146R/K146R} organoids (n= 10 organoids for each group, **Figures 3B** [↗](#), **3D**).

Finally, in order to take into account the possibility of regionalized sampling or damage in the process of histological section, we also confirmed that ACTH production was nearly abolished in *TBX19*^{K146R/K146R} organoids overall on day 105 as compared to control organoids. To make this assessment, whole organoids were rendered completely transparent using a previously validated clearing protocol and imaged by light-sheet microscopy for quantitative analyses (**Figures 4A** [↗](#)-**4B** [↗](#) and **Suppl Movies 1-2**). Indeed, differentiation into corticotroph cells was not uniform throughout the organoids but was concentrated in one or multiple buds. These data demonstrate that the *TBX19*^{K146R/K146R} genotype leads to a significant decrease in ACTH⁺ corticotroph cell numbers by day 105 as compared to control organoids (p= 0.0159, WT n=4, mutant n=5, **Figure 4C** [↗](#)).

Together, this first set of experiments demonstrated that genome editing of hiPSC bearing a homozygous pathogenic variant of *TBX19* prevented their differentiation into ACTH-producing corticotrophs.

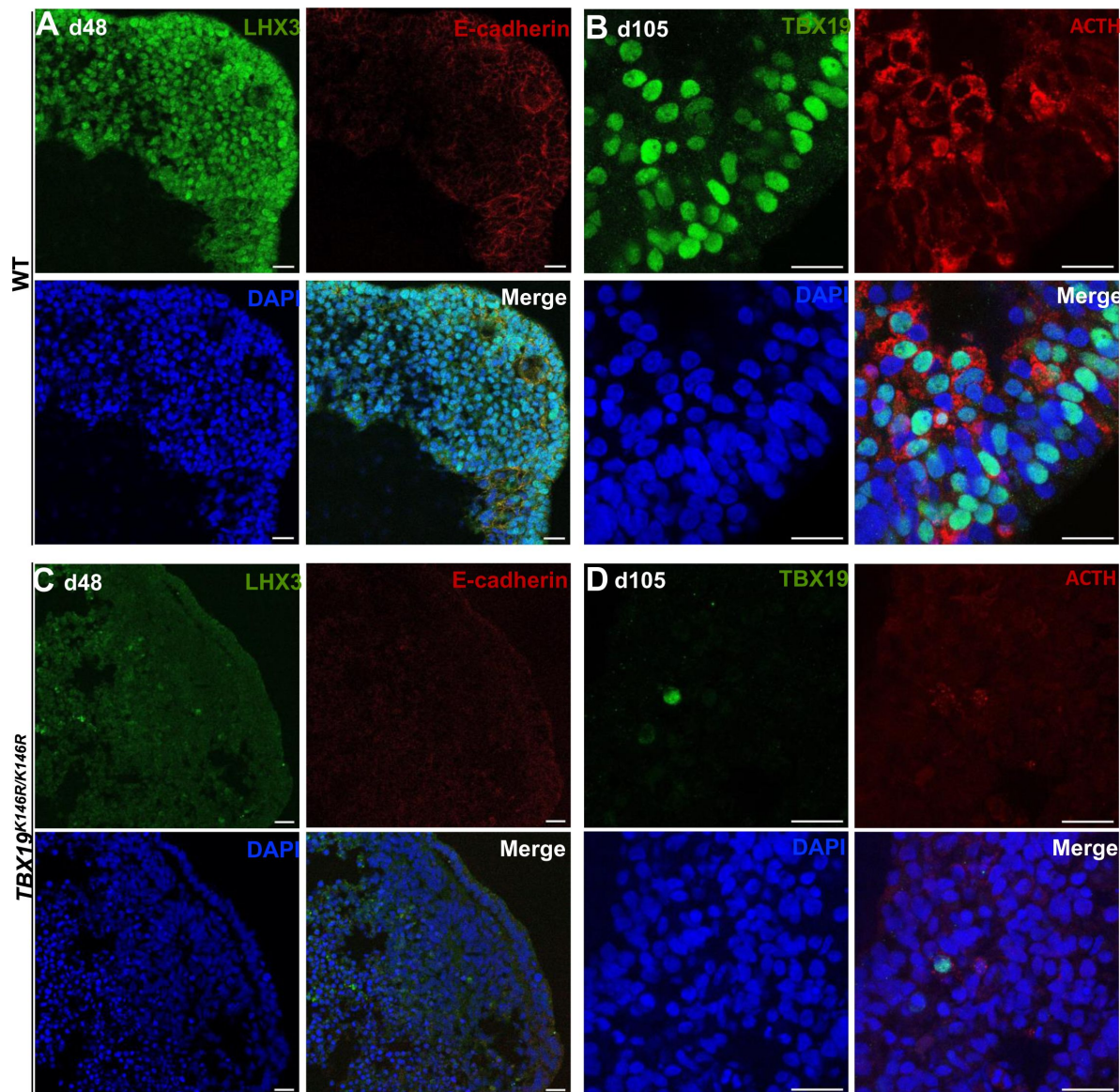


Figure 3

Impairment of corticotroph development in *TBX19^{K146R/K146R}* organoids as compared with control

A, C, Immunostaining of LHX3 and CDH1 (E-cadherin) expression in epithelial cells, typical of Rathke's pouch ectoderm in early pituitary primordia, was reduced in *TBX19^{K146R/K146R}* organoids vs wild-type on day 48 (n=10 organoids for each group). Scale bars : 10 μm.

B, D, Immunostaining showed that ACTH and TBX19 expression was reduced in *TBX19^{K146R/K146R}* organoids vs wild-type on day 105 (n=10 organoids for each group). Scale bars: 10 μm.

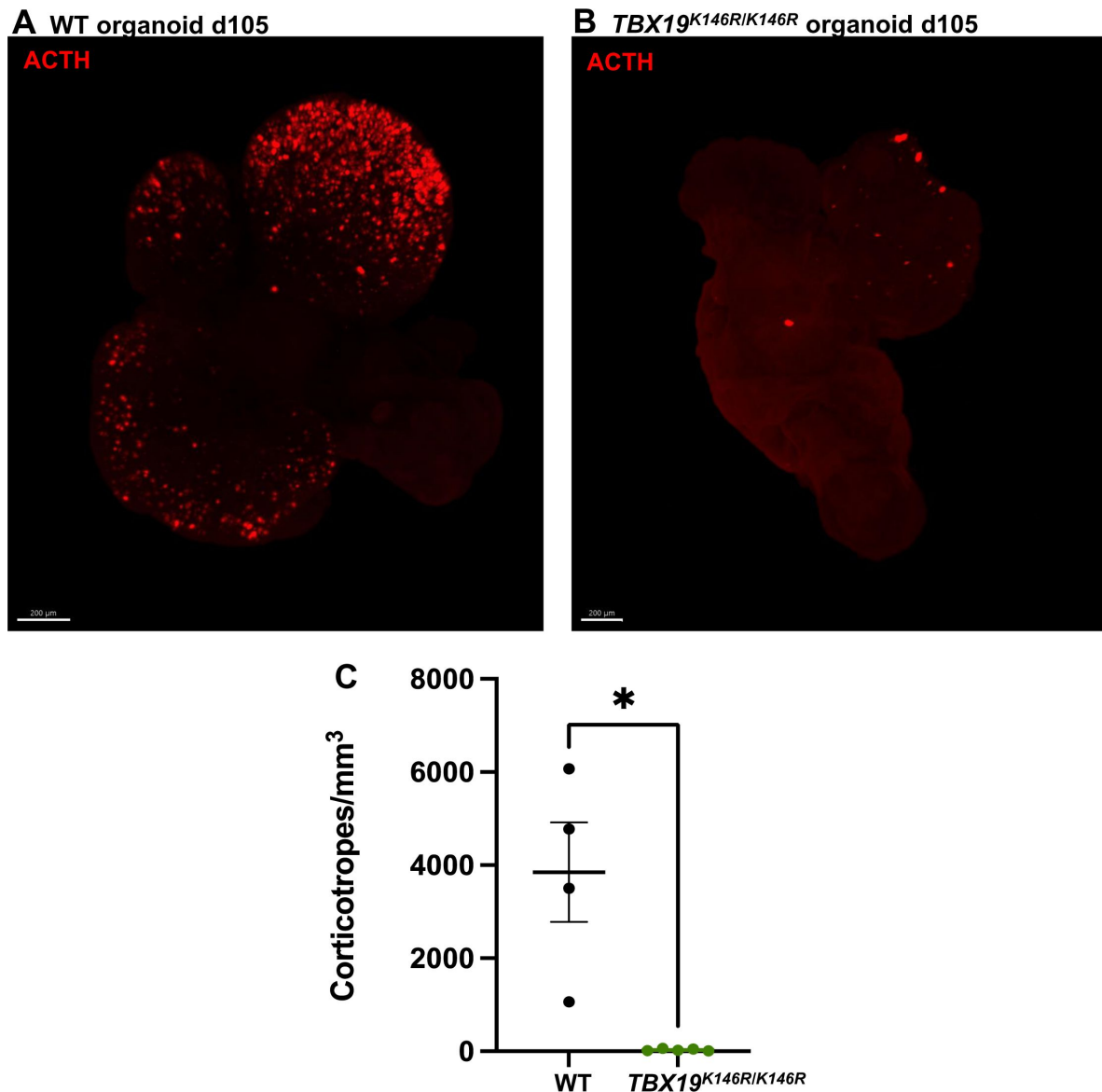


Figure 4

3D reconstruction of whole Wild type (WT) and *TBX19*^{K146R/K146R} organoids on day 105

A, A representative image of whole-mount immunostaining against ACTH in a cleared WT organoid on d105 using light-sheet microscopy. Scale bar: 200 μ m.

B, A representative image of whole-mount immunostaining against ACTH in a cleared *TBX19*^{K146R/K146R} organoid on d105 as above, showing impaired corticotroph differentiation. Scale bar: 200 μ m.

C, The number of corticotroph cells per mm³ was significantly decreased in *TBX19*^{K146R/K146R} organoids (* p = 0.0159). Means $\pm$ SEM for n=4 WT, n=5 *TBX19*^{K146R/K146R} organoids. Mann-Whitney test (unpaired, two-tailed, nonparametric).

NFKB2 signaling is vital for corticotroph development

After validating this organoid model for the study of ACTHD with a known *TBX19* missense mutation, we used the same approach to investigate the potential role of NFKB2 during hypothalamic-pituitary development. The *NFKB2*^{D865G} homozygous mutation was chosen because it was shown to severely affect NFKB2 p52 processing¹². To do so, we first established a *NFKB2*^{D865G/D865G} mutant hiPSC line using CRISPR/Cas9 (**Figure 5**) from the same isogenic control line. Two homozygous KI clones carrying *NFKB2*^{D865G/D865G} were obtained after screening by CAPS and confirmation by Sanger sequencing (**Suppl Figure S4**). RNA-seq analyses confirmed the absence of mutation in the top four predicted off-target sites. We then selected one homozygous missense mutation *NFKB2*^{D865G/D865G} clone (#7) to amplify and generate 200 mutant organoids in parallel with 200 WT organoids in 3D culture.

Next, we compared the development and differentiation of the *NFKB2*^{D865G/D865G} versus WT organoids during culture by qRT-PCR and immunofluorescence as described above. *HESX1* showed no significant difference in expression over time (**Figure 6A**). In contrast, *PITX1* and *LHX3* were significantly decreased in *NFKB2*^{D865G/D865G} organoids on days 27 and 48 (**Figures 6B**–**6C**). Surprisingly, at the latest stage of differentiation (day 105), we observed a significant increase in *TBX19* expression in the mutant group ($p = 0.0095$, WT $n = 4$, mutant $n = 6$, **Figure 6D**). In contrast, *POMC* expression levels were significantly lower in *NFKB2*^{D865G/D865G} organoids compared to WT ($p = 0.019$, **Figure 6E**). Of note, as MRI has revealed pituitary hypoplasia in 43% of patients with DAVID syndrome⁴², we asked whether mutant organoids might be smaller than WT organoids. However, after calculating the volume of organoids on day 105 using Imaris software, these were not significantly different ($p = 0.6$, WT $n = 7$, mutant $n = 8$, **Figure 6F**). Concordant with qRT-PCR results, immunostaining showed less LHX3 expression in *NFKB2*^{D865G/D865G} organoids on day 48 compared to WT (**Figures 7A**–**7B**). Similar results were obtained from ten individual organoids for each line. NFKB2 was co-expressed in LHX3+ pituitary progenitors on day 48 (**Figure 7A**). Unexpectedly, NFKB2 was markedly less expressed in *NFKB2*^{D865G/D865G} organoids than in WT organoids (**Figures 7A**–**7B**). On day 105, immunostaining demonstrated fewer ACTH+ cells; in addition, some TBX19+ cells expressed no simultaneous ACTH signal in *NFKB2*^{D865G/D865G} organoids (**Figures 7C**–**7D**).

Three-dimensional reconstruction of whole organoids highlighted this significantly decreased ACTH expression in *NFKB2*^{D865G/D865G} organoids on day 105 as compared to WT in both qualitative (**Figures 8A**–**8B** and **Suppl Movies 3-4**) and quantitative ($p = 0.0007$, WT $n = 6$, mutant $n = 8$ **Figure 8C**) assessments. In the absence of an immune system, this is strong evidence in support of a direct role for NFKB2 signaling in pituitary differentiation.

To investigate downstream pathways altered in *NFKB2*^{D865G/D865G} pituitary organoids, we performed bulk RNA-seq of five distinct *NFKB2*^{D865G/D865G} organoids versus five organoids derived from its isogenic hiPSC control line on day 48. Differential expression (DE) gene analysis identified 2,829 significantly upregulated and 2,586 significantly downregulated genes at adjusted p ($p_{\text{Adj}} < 0.05$). The global results are depicted as a heatmap to show the similarities across organoids sharing genotypes (**Figure 9A**) and a volcano plot to highlight the magnitudes of DE between the two groups (**Figure 9B**).

Global NFKB signaling was not affected in *NFKB2*^{D865G/D865G} organoids, as most genes assigned to that pathway had fold-change values between -0.5 and 0.5, when they were significant (**Suppl Figure S5**). However, among a list of 143 genes known to have a functional influence on pituitary-hypothalamic development curated from the published literature and also expressed in our model, 66 of these were found to be DE in *NFKB2*^{D865G/D865G} organoids ($p_{\text{Adj}} < 0.05$; **Figure 9B**, **and Table 1**). 42 genes had transcript level changes of two-fold or more and 21 of these were transcription factors (**Figure 9C**). Expression of most of these transcription factors was downregulated, with a marked decrease in expression of anterior pituitary progenitor markers such as *PITX1* and *LHX3*, as well as lineage precursor markers *PROP1* and *POU1F1* (**Figure 9C**).

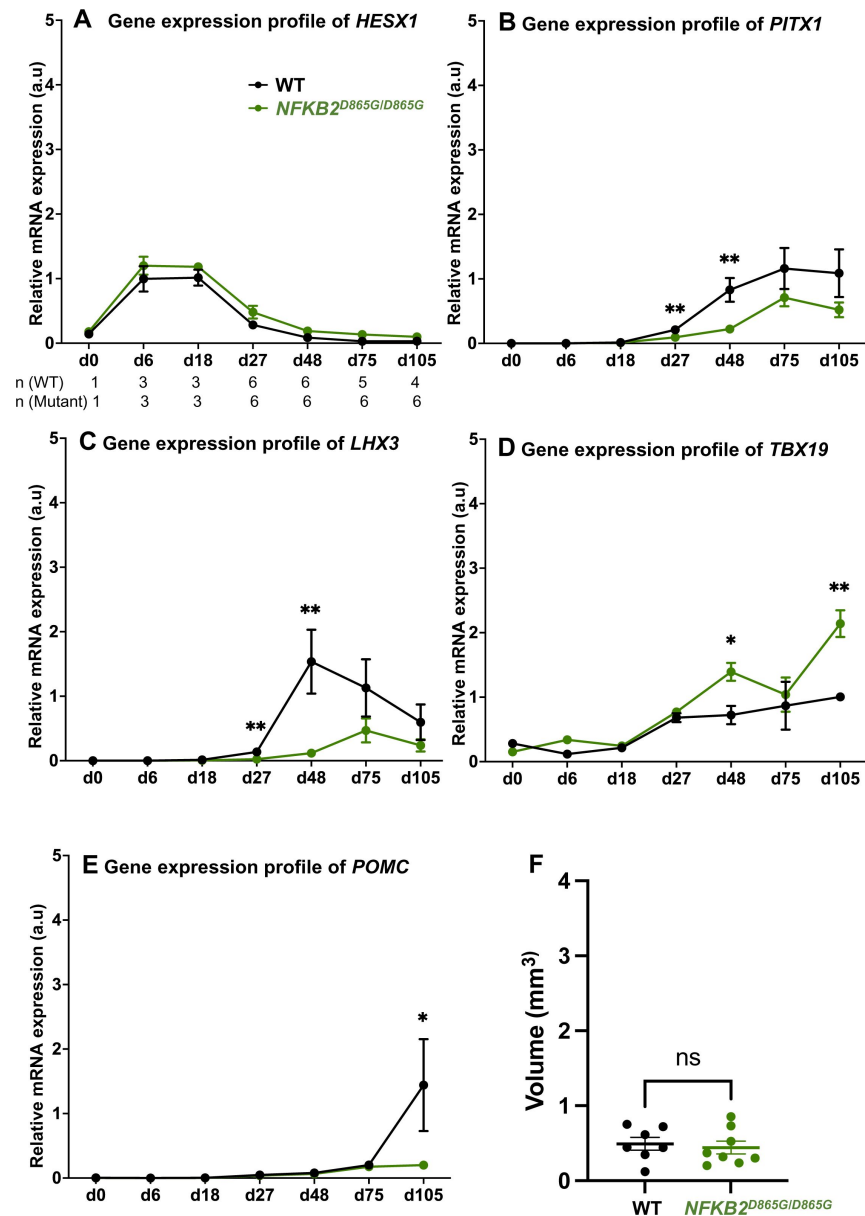


Figure 6

Time course of organoid growth and gene expression in WT and *NFKB2^{D865G/D865G}* organoids

A-E, Relative quantification (RQ) mRNA expression analysis for key markers of pituitary organoids during differentiation: WT (in black line) and *NFKB2^{D865G/D865G}* organoids (in green line). Data show means $\pm$ standard error of the mean (SEM); n=1 on day (d)0, n=3 samples per group (8 organoids per sample) at the time points d6 and d18, n=4-6 organoids per group at each following time point (d27, d48, d75 and d105). Mann-Whitney t-test (unpaired, two-tailed, nonparametric). $p < 0.05$ (*), $p < 0.01$ (**).

A, Relative quantification expression of *HESX1*. The expression of *HESX1* was not significantly different in *NFKB2^{D865G/D865G}* organoids vs. WT.

B, *PITX1* was significantly downregulated in *NFKB2^{D865G/D865G}* organoids by d27 (** $p = 0.0022$) and d48 (* $p = 0.0022$).

C, *LHX3* was significantly lower in *NFKB2^{D865G/D865G}* organoids as compared to WT on d27 (** $p = 0.0087$), and d48 (** $p = 0.0087$).

D, *TBX19* was significantly increased in *NFKB2^{D865G/D865G}* organoids by d48 (* $p = 0.026$) and d105 ($p = 0.0095$).

E, *POMC* was significantly downregulated in *NFKB2^{D865G/D865G}* organoids by d105 (* $p = 0.019$).

F, There was no significant difference in volume between WT and *NFKB2^{D865G/D865G}* organoids ($p = 0.6126$). Data show means $\pm$ SEM; n=7 in the WT group, n=8 in the mutant group. Mann-Whitney test (unpaired, two-tailed).

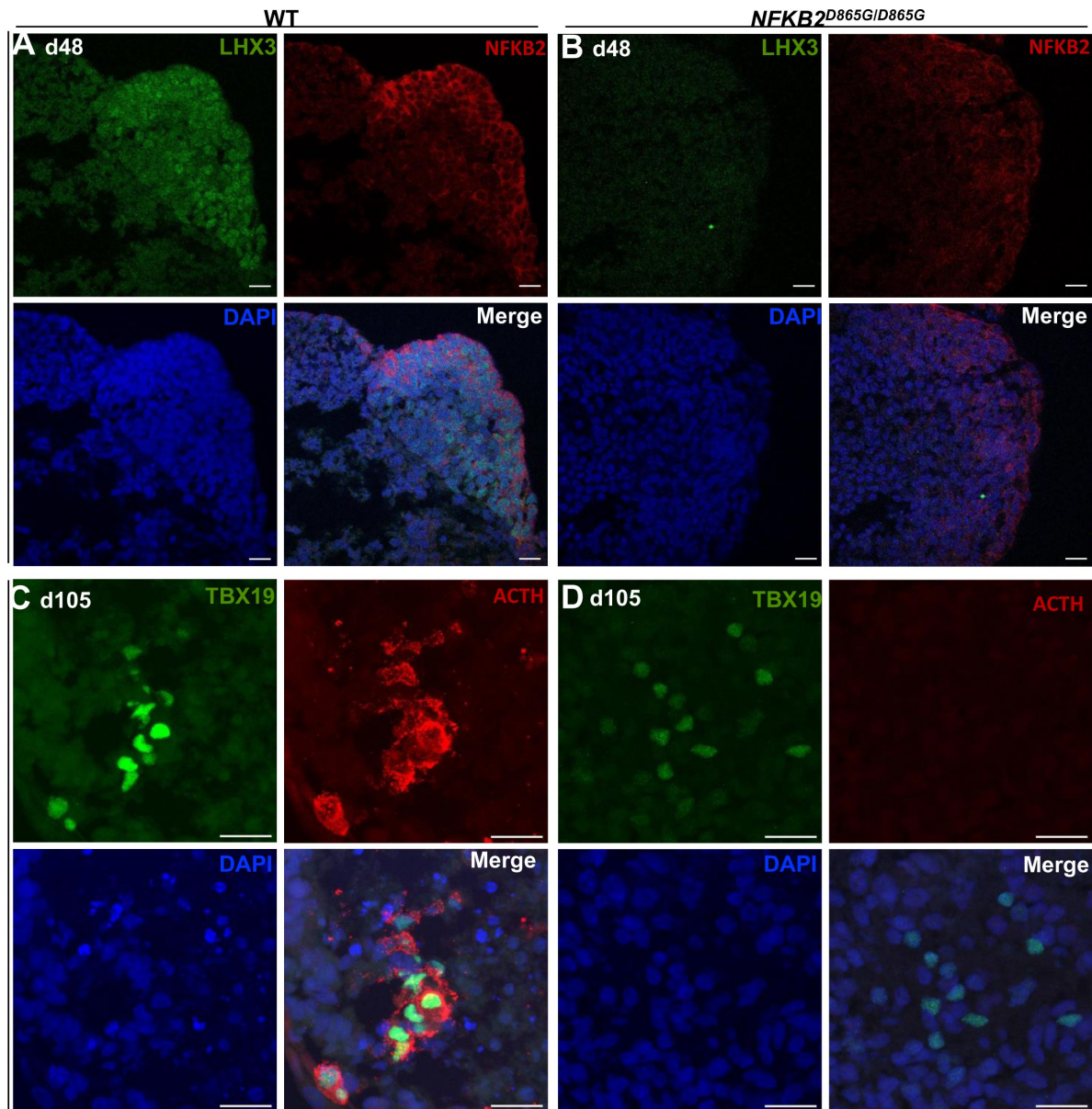


Figure 7

Impairment of pituitary development in *NFKB2^{D865G/D865G}* organoids

A, B, Immunostaining of LHX3 and NFKB2 expression in the early pituitary-type epithelium was reduced in *NFKB2^{D865G/D865G}* organoids vs WT on day 48 (n=10 organoids for each group). Scale bars: 10 μ m

C, D, Immunostaining showed that by day 105, there was complete absence of corticotroph cells co-expressing ACTH and TBX19 in *NFKB2^{D865G/D865G}* organoids versus WT organoids on day 105 (n=10 organoids for each group). Note the population of cells expressing TBX19 without ACTH in the mutant example. Scale bars: 10 μ m

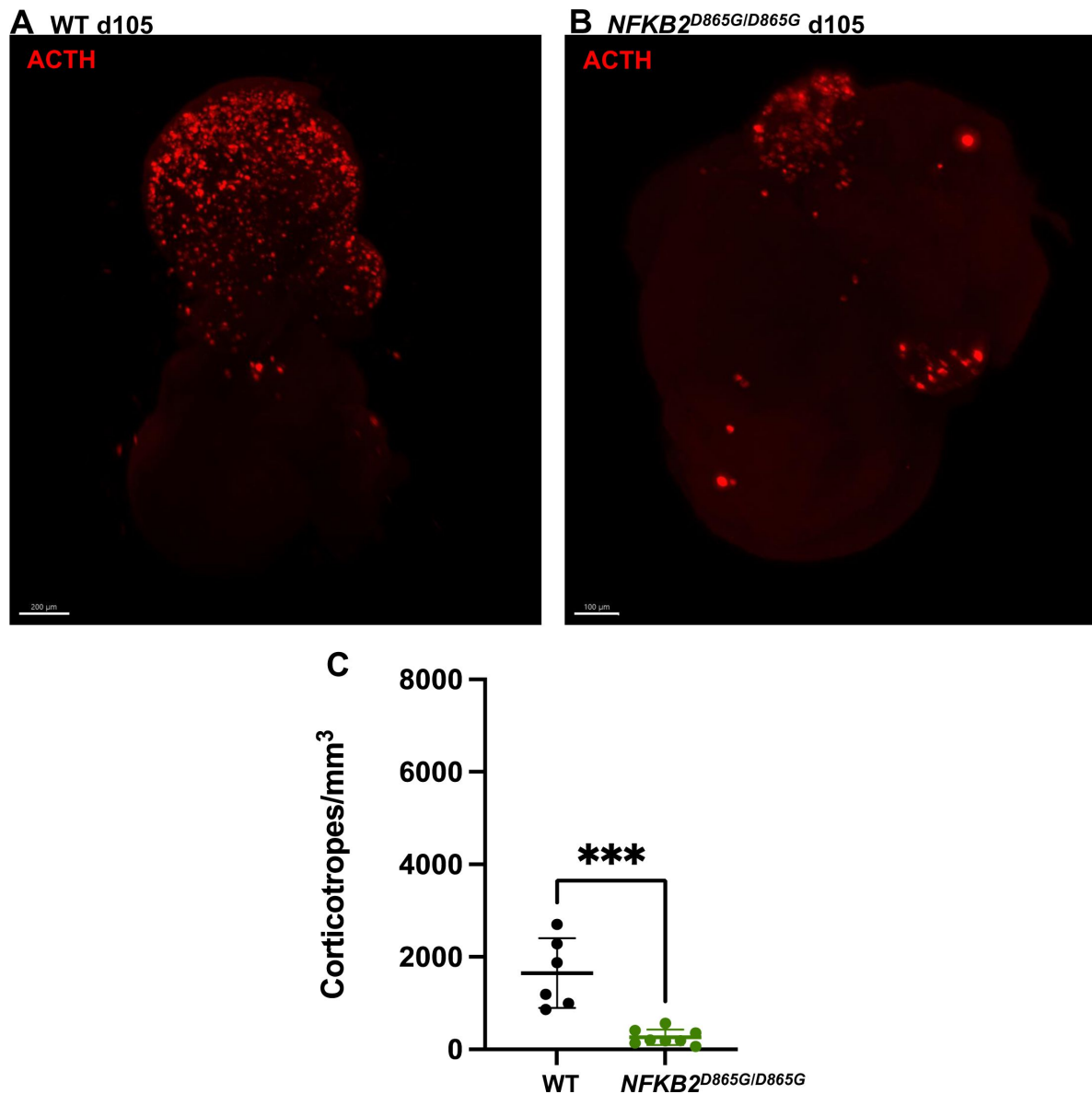


Figure 8

3D reconstruction of whole *NFKB2*^{D865G/D865G} or WT organoids on day 105

A, A representative image of whole-mount immunostaining against ACTH in a cleared WT organoid on d105 using light-sheet microscopy. Scale bar: 200 μ m.

B, A representative image of whole-mount immunostaining against ACTH in a cleared *NFKB2*^{D865G/D865G} organoid on d105 as above, showing impaired corticotroph differentiation. Scale bar: 100 μ m.

C, The number of corticotroph cells per mm³ was significantly decreased in *NFKB2*^{D865G/D865G} organoids (* $p = 0.0007$). Means $\pm$ SEM for $n = 6$ WT organoids, $n = 8$ *NFKB2*^{D865G/D865G} organoids. Mann-Whitney test (unpaired, two-tailed, nonparametric).

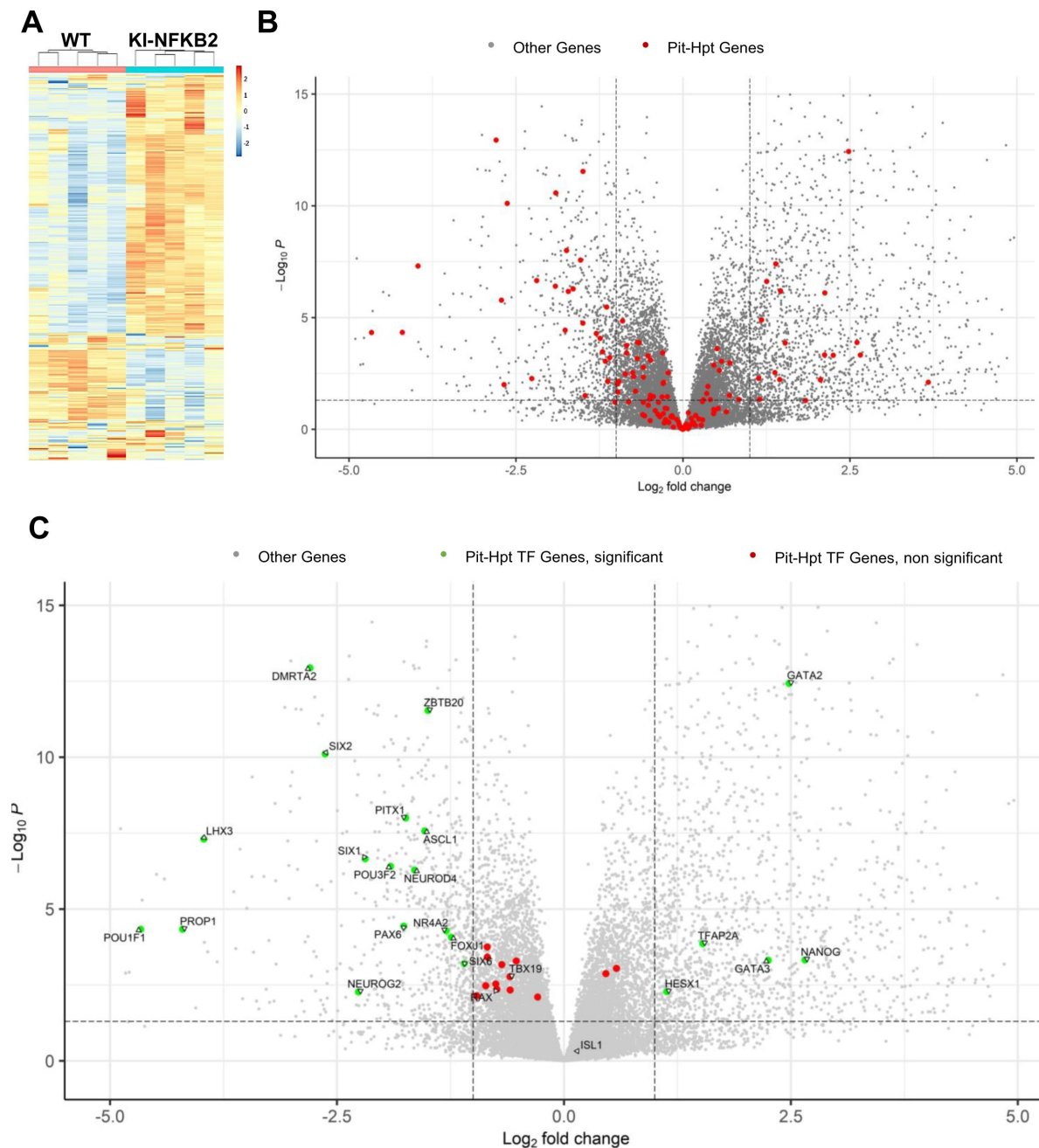


Figure 9

Whole transcriptome profiles of WT vs *NFKB2^{D865G/D865G}* organoids on day 48

A, Heat map showing global differential gene expression between WT and *NFKB2^{D865G/D865G}* organoids (n=5 for each group). **B**, Volcano plot of genes showing differential gene expression between WT and *NFKB2^{D865G/D865G}* organoids. Red dots in the left panel indicate 143 genes involved in pituitary-hypothalamic development. Grey dots indicate all other genes detected in RNA-seq.

C, Volcano plot of genes showing differential gene expression between WT vs *NFKB2^{D865G/D865G}* organoids. Green dots indicate genes coding for 21 key transcription factors (TF) whose expression is significantly changed (fold change < 0.5 or > 2, padj < 0.05). Red dots indicate fold change in expression between 0.5 and 2. Grey dots indicate all other genes detected in RNA-seq.

Pit-Hpt: pituitary-hypothalamic; KI: knock-in; TF: transcription factors

and Table 1). In contrast, the earliest Rathke's pouch marker, *HESX1*, showed a 2-fold increased expression in *NFKB2*^{D865G/D865G} organoids, suggesting an impaired progression of pituitary progenitors towards more differentiated stages, with increased expression of epithelial markers *CDH1*, *KRT8* and *CLDN6* and decreased transcription of mesenchymal markers *CDH2* and *VIM* supporting that hypothesis (Figure 9C and Table 1).

The transcription factor *RAX* also showed a moderate but significant decrease in its expression (fold change = -0.6, *p*adj < 0.05) in *NFKB2*^{D865G/D865G} organoids (Figure 9C and Table 1), suggesting that the phenotype could be due, at least partially, to a hypothalamic defect. Indeed, among the factors known to mediate the interaction between the hypothalamus and oral ectoderm during pituitary development, *BMP4* and *FGF10* had significant fold-changes of 4.35 and 0.26 in *NFKB2*^{D865G/D865G} organoids respectively, whereas *SHH*, *WNT5A* and *FGF8* expression levels were not modified (Table 1). Finally, a 2.4-fold increase in *FST* expression, a gene recently identified as a marker of differentiating corticotrophs¹⁶, and concomitantly decreased levels of mature corticotroph markers *AR*, *NR4A2*, *PCSK1* and *POMC* (0.63, 0.4, 0.35 and 0.43-fold changes respectively) (Table 1), corroborate the hypothesis that TBX19-positive precursors fail to achieve terminal differentiation in *NFKB2*^{D865G/D865G} organoids.

Taken together, our data identify NFKB2-mediated signaling as an important factor acting on pituitary development at multiple levels during its maturation.

Discussion

After describing the rare combination of an anterior pituitary deficit, mostly ACTHD, with common variable immunodeficiency in DAVID syndrome, we and others found that affected patients carry *NFKB2* gene mutations affecting specific C-terminal residues of the NFKB2 protein known to play important roles in immunity^{7,9}. As the spontaneous mouse knockout mutant of the orthologous gene lacks an endocrine phenotype, yet *NFKB2* is transcribed in the prenatal human pituitary gland, we wanted to investigate the mechanisms of ACTHD in a human disease model.

The available *in vitro* models to study ACTHD and CPHD had strong limitations. For instance, studies based on transfections and luciferase-coupled hormone promoters can only give indirect evidence of the effect of a variant of unknown significance on the activity of a given transcription factor, as they are based on non-physiological amounts of DNA and promoter constructs⁵. Murine models only partially replicate human CPHD, as already shown for *PROT1* mutations in ACTHD (absent in mice but present in 40% of humans)⁴³. However, reprogramming differentiated adult cells into induced pluripotent stem cells and advances in genome editing technologies using CRISPR/Cas9 broadened possibilities for modeling novel aspects of human disease *ex vivo*⁴⁴. Organoid culture has allowed the modeling of otherwise inaccessible congenital human disorders affecting the brain²⁵, intestine⁴⁵, kidney⁴⁶, liver⁴⁷, pancreas⁴⁸, ovary⁴⁹, and lung⁵⁰. In the present study, we established a human *in vitro* model of congenital ACTHD using 3D pituitary organoids differentiated from *TBX19*^{K146R/K146R} and *NFKB2*^{D865G/D865G} hiPSC generated by CRISPR/Cas9 genome editing, in order to compare them with their isogenic WT equivalents. After confirming that this model could recapitulate relevant aspects of pituitary development, we then determined whether it could also replicate a disease, namely ACTHD.

As the *TBX19*^{K146R} homozygous variant is known to cause ACTHD in humans, we designed a human organoid model derived from *TBX19*^{K146R/K146R} hiPSC as a proof of principle. In line with the *Tbx19*^{-/-} mouse model³, *TBX19*^{K146R/K146R} organoids displayed markedly defective corticotroph differentiation, as compared to WT organoids, thereby validating the methodological approach. Interestingly, in organoids carrying a *TBX19*^{K146R/K146R} missense mutation that affects

gene_name	FoldChange	padj	gene_name	FoldChange	padj	gene_name	FoldChange	padj
ADCYAP1	0,447	0,007888035	GNRHR	0,453	8,31212E-05	POLR3A	1,014	0,907184037
AIP	0,898	0,534416537	GNRHR2	0,698	0,239376773	POMC	0,434	0,003599837
ALDH1A2	0,152	4,59754E-05	GPR101	1,209	0,830451145	POU1F1	0,039	0,000715204
ALDH1A3	0,182	2,61E-13	GREB1	1,007	0,977843175	POU3F1	1,374	0,420928494
AR	0,634	0,001676979	GSX1	0,659	0,450115297	POU3F2	0,266	1,36281E-05
ARNT2	0,534	0,000269934	HES1	0,663	0,028047242	POU5F1	3,573	0,170177475
ASCL1	0,345	1,44E-06	HESX1	2,193	0,031209258	PROKR2	1,056	0,973652742
AXIN2	1,296	0,058892068	HEY1	0,804	0,130459101	PROP1	0,054	0,000712172
BMP2	0,750	0,342780888	HMG2	0,775	0,194708366	RAX	0,599	0,026972637
BMP4	4,357	2,4368E-05	INHBB	0,458	0,039744399	RBPJ	0,814	0,046649215
BMPR1A	1,019	0,919913721	ISL1	1,097	0,758119581	S100B	0,304	2,15503E-05
BRAF	0,892	0,481826746	KDM1A	1,021	0,903369765	SALL1	1,106	0,761223759
CDH1	1,626	0,008999389	KRT8	2,617	1,99E-06	SHH	0,786	0,511785531
CGA	12,756	0,042344376	LATS1	0,857	0,019968707	SIX1	0,219	8,42E-06
CREB1	0,850	0,129014692	LATS2	1,460	0,016632236	SIX2	0,161	8,97E-09
CRH	4,347	0,004688935	LHX2	0,622	0,006247085	SIX3	0,859	0,699762446
CRHR1	0,156	0,051309993	LHX3	0,064	2,39E-06	SIX6	0,467	0,005859008
CRHR2	0,823	0,736016163	LHX4	0,557	0,003977929	SLC15A2	0,624	0,001635121
CTNNB1	1,009	0,955419302	MEN1	1,030	0,803086498	SLC6A3	2,215	0,157967352
CYP11A1	6,086	0,001655774	MSX1	0,904	0,909088208	SMO	0,957	0,7040106
DIO2	1,114	0,718852455	NANOG	6,301	0,004723205	SMOC2	4,163	0,034876305
DISP1	1,142	0,477079028	NEUROD1	0,883	0,509660164	SOX11	0,736	0,127568673
DLK1	0,594	0,019880505	NEUROD2	1,137	0,841778563	SOX2	0,815	0,421505197
DMRTA2	0,144	3,58E-11	NEUROD4	0,320	1,72108E-05	SOX3	0,787	0,501517696
DRD2	2,753	2,04541E-05	NEUROG1	0,493	0,192091391	SOX9	0,557	0,002135493
DUOX2	2,595	0,019613055	NEUROG2	0,208	0,031653158	SRD5A1	0,961	0,79794316
DZIP1	0,818	0,042643465	NFKB2	1,236	0,161439179	SST	0,994	0,992100441
EGR1	1,784	0,157681046	NKX2-1	0,707	0,15517589	SSTR1	0,509	0,090081721
ETS1	3,947	4,82E-13	NKX2-2	1,569	0,373223947	SSTR2	0,769	0,435993359
EYA1	0,514	0,039620927	NOG	1,376	0,318972037	TAZ	0,967	0,853180694
FGF10	0,267	3,49E-09	NOTCH1	1,006	0,98289259	TBX19	0,663	0,012939434
FGF2	1,224	0,182371108	NOTCH2	0,937	0,606914403	TBX3	0,997	0,989105106
FGF3	0,611	0,084446193	NR0B1	1,062	0,928929659	TCF4	1,015	0,957689002
FGF8	0,506	0,048269157	NR3C1	0,846	0,302794383	TCF7L2	1,178	0,588438732
FOLR1	2,729	0,034433598	NR4A1	1,227	0,621258008	TFAP2A	2,887	0,001732632
FOXA1	0,550	0,02192538	NR4A2	0,407	0,000780424	TGIF1	1,422	0,002776353
FOXC1	1,619	0,118175086	NR5A1	0,834	0,613961749	THRA	0,714	0,007115882
FOXJ1	0,424	0,001168862	NUPR1	1,384	0,347695524	THRB	2,260	0,000249134
FOXO1	1,494	0,007821304	OLFM1	0,830	0,291658503	TIPRL	0,810	0,003930719
FST	2,384	9,05E-06	OTP	0,663	0,193485664	TLE5	0,695	0,005015424
GATA2	5,579	9,69E-11	OTX1	0,863	0,722257262	TSHR	0,998	0,997529133
GATA3	4,758	0,004829173	OTX2	0,976	0,94712869	USP39	1,035	0,790802941
GHRH	0,712	0,650531144	PAX3	0,362	0,119629185	WNT4	0,712	0,116649738
GHRHR	0,672	0,485122167	PAX6	0,294	0,000592101	WNT5A	1,442	0,295426855
GLI1	1,050	0,841041925	PAX7	0,570	0,188199017	YAP1	1,091	0,565853991
GLI2	1,328	0,159304929	PCSK1	0,354	0,000327177	ZBTB20	0,354	5,29E-10
GLI3	1,279	0,10094389	PITX1	0,299	6,27E-07	ZEB2	1,378	0,010775016
GNAS	1,058	0,395187491	PITX2	1,072	0,86688154			

Table 1

RNA-seq expression data for a list of 143 genes known to have a functional influence on pituitary-hypothalamic development. Differentially expressed genes (padj < 0.05) in *NFKB2^{D865G/D865G}* organoids are highlighted in green.

DNA binding, a few corticotroph cells were still present, suggesting either a role for TBX19 independent of its DNA binding activity or persistence of residual DNA binding activity with this mutation.

Using this same approach with *NFKB2*^{D865G/D865G} organoids generated by CRISPR/Cas9-edited hiPSC, we demonstrated for the first time a role for NFKB2 in pituitary development. Heterozygous mutations in the C-terminal region of *NFKB2*, such as D865G, were reported in DAVID syndrome, leading to the disruption of both non-canonical and canonical pathways^{8, 11–13, 51, 52}. Although *NFKB2*^{D865G} has been identified in a heterozygous state in patients with ACTHD, we decided to generate a homozygous model of *NFKB2*^{D865G/D865G} to determine first whether NFKB2 was involved in human pituitary development at all and could explain the defect in ACTH production in DAVID syndrome (OMIM#, 615577)^{7, 8, 53}. Indeed, while concomitant common variable immune deficiency (CVID) can be explained by the key roles of NFKB signaling in the immune system, the mechanism of the endocrine deficits caused by *NFKB2* mutations was unknown. In mouse models, deletion of *Nfkb2* causes abnormal germinal center B-cell formation and differentiation¹⁴. However, the spontaneous *Lym1* mouse model, which carries a truncating variant Y868* in the *Nfkb2* orthologue that prevents its cleavage into a transcriptionally active form, had apparently normal corticotroph function and ACTH expression in the pituitary in both heterozygotes and homozygotes⁹. Mutant *Nfkb2* homozygotes do display reduced fertility and more severe immune abnormalities than their heterozygous counterparts⁵⁴. Using pituitary organoids differentiated from a *NFKB2*^{D865G/D865G} hiPSC line in comparison with its unmutated control and in the absence of an immune response, we have demonstrated that human corticotroph development is directly disrupted by the same missense mutation found in DAVID syndrome.

The precise molecular mechanisms by which NFKB2 impairs corticotroph development remain to be determined. However, our results suggest both early and late effects on corticotroph differentiation. First, NFKB2 may be involved in the early steps of pituitary development. RNA-seq data on day 48 of *NFKB2*^{D865G/D865G} organoid development showed increased *HESX1* and *BMP4* transcription, which may reflect blockade in an undifferentiated state evocative of the Rathke's pouch epithelial primordium: indeed, down-regulation of *HESX1* is necessary for proper pituitary development⁵⁵. This is also supported by RNA-seq results showing downregulation of *PITX1*, *HES1*, *LHX3* and *FGF10* in mutant organoids, as their expression is known to be necessary for progression to a later progenitor state^{56, 57}. Finally, concordant with the hypothesis of a differentiation blockade, upregulation of epithelial markers such as E-cadherin and keratin-8 was observed in mutant *NFKB2*^{D865G/D865G} organoids on day 48 while in contrast, there were no changes in Wnt and hedgehog signaling pathways. Second, NFKB2 could also be involved in the final steps of corticotroph differentiation. While qRT-PCR showed increased levels of *TBX19* mRNA in mutant organoids, this did not lead to the expected increase in *POMC* mRNA transcripts. Similarly, some *NFKB2* mutants had TBX19+ cells without simultaneous ACTH expression. This suggests that NFKB2 is necessary for *POMC* synthesis in TBX19-specified cells, which *NFKB2*^{D865G} may prevent. This could be due to abnormal DNA binding of the NFKB heterodimer to the *POMC* promoter^{58, 59} and/or regulation of *PCSK1*, the gene coding for the enzyme necessary to convert *POMC* to ACTH. Alternatively, *NFKB2*^{D865G} could also block TBX19+ cells in a pre-differentiated corticotroph state through its derepression of chromatin remodeling factors such as *EZH2*⁶⁰.

The organoid model described in this paper presents several advantages over other approaches. It is a true human model of ACTHD, emphasizing the limits of mouse models as exemplified by the *Lym1* model, which showed that *Nfkb2* was dispensable for murine pituitary development. Despite some technical constraints, CRISPR/Cas9-based genome editing of hiPSC is a powerful approach to elucidate gene function in congenital pituitary deficiency patients, as shown for the role of hypothalamic OTX2 regulation of pituitary progenitor cells in congenital pituitary hypoplasia²⁰. While the prevalence of pathogenic variants of known genes in primary

hypopituitarism is currently estimated to be below 10%⁶⁰, exome and whole-genome sequencing regularly identify new genes and variants, for which pathogenicity remains uncertain³⁵. This model or refinements thereof could thus become the gold standard to confirm involvement of a given gene in human anterior pituitary development, or to classify new variants of unknown significance as likely benign or pathogenic. Finally, pituitary organoids derived from human embryonic stem cells have been transplanted subcutaneously into mice with hypopituitarism; grafted cells were able to secrete ACTH and respond to corticotropin-releasing hormone stimulation⁶¹. The near future may therefore promise patients with hypopituitarism innovative substitutive treatments with their own reprogrammed cells.

In conclusion, we have designed and validated a model replicating human constitutive ACTHD using a combination of CRISPR/Cas9 and directed differentiation of hiPSC into 3D hypothalamic-pituitary organoids. Not only has this proof-of-concept reproduced the known prerequisite for the TBX19 transcription factor in human corticotroph differentiation, it has also allowed a new classification of a *NFKB2* variant of previously unknown clinical significance as pathogenic in pituitary development. From a clinical viewpoint, our organoid model can provide evidence for the pathogenic role of new candidate genes or variants investigated in patients where none other is available, especially in a rare disease affecting a poorly accessible organ like the pituitary gland. From a physiological viewpoint, modeling corticotroph deficiency using hiPSC allows to support the functional role of *NFKB2* in human pituitary differentiation.

Acknowledgements

This study was supported by the ADEREM (Association pour le Développement de la Recherche Médicale au CHU de Marseille), the Association Française contre les Myopathies (AFM-Téléthon) MoThARD program and by Pfizer. TTM received funding from France Excellence Scholarship of the French Embassy in Vietnam, from the French Society of Pediatric Endocrinology and Diabetology (SFEDP) and from the Marseille Rare Disease (MarMaRa) Institute of Aix-Marseille University. The authors wish to thank Frederique Magdinier, Natacha Broucqsault and Claire El Yazidi from the Marseille Medical Genetics (MMG) cell reprogramming & differentiation facility (MaSC); Sébastien Courrier, from the MMG microscopy platform; Valerie Delague, Catherine Aubert, Christel Castro and Camille Humbert from MMG's Genomics and Bioinformatics (GBiM) platform; Carole Siret and Mathieu Fallet from the Centre d'Immunologie de Marseille-Luminy (CIML); Ivo Vanzetta and Alberto Lombardini from the Institut de Neurosciences de la Timone (INT) microscopy platform, and Laurie Arnaud and Emmanuel Nivet from the Institut de Neuropsychiatrie (INP), all in Marseille, for their expert assistance, advice or reagents.

Declaration of interests

The authors declare no competing interests.

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Editors

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Reviewer #1 (Public Review):

Summary:

NFKB mutations are thought to be one of the causes of pituitary dysfunction, but until now they could not be reproduced in mice and their pathomechanism was unknown. The authors used the differentiation of hypothalamic-pituitary organoids from human pluripotent stem cells to recapitulate the disease in human iPSCs carrying the NFKB mutation.

Strengths:

The authors achieved their primary goal of recapitulating the disease in human cells. In particular, the differentiation of the pituitary gland is closely linked to the adjacent hypothalamus in embryology, and the authors have again shown that this method is useful when the hypothalamus is suspected to be involved in pituitary abnormalities caused by genetic mutations.

Weaknesses:

On the other hand, the pathomechanism is still not fully understood. This study provides some clues to the pathomechanism, but further analysis of NFKB expression and experiments investigating the relevant factors in more detail may help to clarify it further.

Reviewer #2 (Public Review):**Summary:**

DAVID syndrome is a rare autosomal dominant disorder characterized by variable immune dysfunction and variable ACTH deficiency. Nine different families have been reported, and all have heterozygous mutations in NFKB2. The mechanism of NFKB2 action in the immune systems has been well-studied, but nothing is known about its role in the pituitary gland.

The DAVID mutations cluster in the C-terminus of the NFKB2 and interfere with cleavage and nuclear translocation. The mutations are likely dominant negative, by affecting dimer function. ACTH deficiency can be life-threatening in neonates and adults, thus, understanding the mechanism of NFKB2 action in pituitary development and/or function is important.

The authors use CRISPR/Cas gene editing of human iPSC-derived pituitary-hypothalamic organoids to assess the function of NFKB2 and TBX19 in pituitary development. Mutations in TBX19 are the most common, known cause of pituitary ACTH deficiency, and the mechanism of action has been studied in mice, which phenocopy the human condition. Thus, the TBX19 organoids can serve as a positive control. The Nfkb2 mouse model has a p.Y868* mutation that impairs cleavage of NFKB2 p100, and the immune phenotype mimics the patients with DAVID mutations, but no pituitary phenotype was evident. Thus, a human organoid model might be the only approach suitable to discover the etiology of the pituitary phenotype.

Overall, the authors have selected an important problem, and the results suggest that the pituitary insufficiency in DAVID syndrome is caused by a developmental defect rather than an autoimmune hypophysitis condition. The use of gene editing in human iPSC-derived hypothalamic-pituitary organoids is significant, as there is only one example of this previously, namely studies on OTX2. Only a few laboratories have demonstrated the ability to differentiate iPSC or ES cells to these organoids, and the authors have improved the efficiency of differentiation, which is also significant.

The strength of the evidence is excellent. However, the two ACTH-deficient organoid models use a single genetically engineered clone, and the potential for variability amongst clones makes the conclusions less compelling. Since the authors obtained two independent clones for NFKB2 it is not clear why only one clone was studied. Finally, the effect of TBX19 on early pituitary fate markers is somewhat surprising given the phenotype of the knockout mice and patients with mutations. Thus, the use of a single clone for that study is also worrisome.

Strengths:

The authors make mutations in TBX19 and NFKB2 that exist in affected patients. The TBX19 p.K146R mutation is recessive and causes isolated ACTH deficiency. Mutations in this gene account for 2/3 of isolated ACTH deficiency cases. The NFKB2 p.D865G mutation is heterozygous in a patient with recurrent infections and isolated ACTH deficiency. NFKB2 mutations are a rare cause of ACTH deficiency, and they can be associated with the loss of other pituitary hormones in some cases. However, all reported cases are heterozygous.

The developmental studies of organoid differentiation seem rigorous in that 200 organoids were generated for each hiPSC line, and 3-10 organoids were analyzed for each time point and genotype. Differentiation analysis relied on both RNA transcript measurements and immunohistochemistry of cleared organoids using light sheet microscopy. Multiple time points were examined, including seven times for gene expression at the RNA level and two times in the later stages of differentiation for IHC.

TBX19 deficient organoids exhibit reduced levels of PITX1, LHX3, and POMC (ACTH precursor) expression at the RNA and IHC level, and there are fewer corticotropes in the organoids, as ascertained by POMC IHC.

The NFKB2 deficient organoids have a normal expression of the early pituitary transcription factor HESX1, but reduced expression of PITX2, LHX3, and POMC. Because there is no immune component in the organoid, this shows that NFKB2 mutations can affect corticotrope differentiation to produce POMC. RNA sequencing analysis of the organoids reveals potential downstream targets of NFKB2 action, including a potential effect on epithelial-to-mesenchymal-like transition and selected pituitary and hypothalamic transcription factors and signaling pathways.

Weaknesses:

There could be variation between individual iPSC lines that is unrelated to the genetically engineered change. While the authors check for off-target effects of the guide RNA at predicted sites using WGS, a better control would be to have independently engineered clones or to correct the engineered clone to wild type and show that the phenotypic effects are reversed.

All NFKB2 patients are heterozygous for what appear to be dominant negative mutations that affect protein cleavage and nuclear localization of processed protein as homo or heterodimers. The organoids are homozygous for this mutation. Supplemental Figure 4 indicates that one heterozygous clone and two homozygous mutant clones were obtained. Analysis of these additional clones would give more strength to the conclusions, showing reproducibility and the effect of mutant gene dosage.

Reviewer #3 (Public Review):

Summary:

This manuscript by Mac et al addresses the causes of pituitary dysfunction in patients with DAVID syndrome which is caused by mutations in the NFKB2 gene and leads to ACTH deficiency. The authors seek to determine whether the mutation directly leads to altered pituitary development, as opposed to an autoimmune defect, by using mutating human iPSCs and then establishing organoids that differentiate into pituitary tissue. They first seek to validate the system using a well-characterised mutation of the transcription factor TBX19, which also results in ACTH deficiency in patients. Then they characterise altered pituitary cell differentiation in mutant NFKB2 organoids and show that these lack corticotrophs, which would lead to ACTH deficiency.

Strengths:

The conclusion of the paper that ACTH deficiency in DAVID syndrome is independent of an autoimmune input is strong.

Weaknesses:

1. The authors correctly emphasise the importance of establishing the validity of an iPSC-based model in being able to recapitulate in vivo dysfunctional pituitary development through characterisation of a TBX19 knock-in mutation. Whilst this leads to the expected failure of functional corticotroph differentiation, other aspects of the normal pituitary differentiation pathway upstream of corticotroph commitment seem to have been affected in surprising ways. In particular, the loss of LHX3 and PITX1 in TBX19 mutant organoids

compared with wild type requires explanation, especially as the mutant protein would only be expected to be expressed in a small proportion of anterior pituitary lineage cells. If the developmental expression profile of key transcription factors in mutant organoids does not recapitulate that which occurs *in vivo*, any interpretation of the relevance of expression differences in the NFKB2 organoids to the mechanism(s) leading to corticotroph function *in vivo* has to be questionable. It is notable that the manipulation of iPSC cells used to generate mutants through CRISPR/Cas9 editing is not applied to the control iPSC line. It is possible that these manipulations lead to changes to the iPSC cells that are independent of the mutations introduced and this may change the phenotype of the cells. A better control would have been an iPSC line with a benign knock-in (such as GFP into the ROSA26 locus).

2. In the results section of the manuscript the authors acknowledge that hypothalamic tissue in the NFKB2 mutant organoid may be having an effect on the development of pituitary tissue. However, in the discussion the emphasis is entirely on pituitary autonomous mechanisms such as pituitary HESX1 expression or POMC gene regulation; in the conclusion of the abstract, a direct role for NFKB2 in pituitary differentiation is described. Whilst the data here may suggest a non-immune mediated alteration in pituitary function in DAVID syndrome, if this is due to alteration of the developing hypothalamus then this is not direct. A fuller discussion of the potential hypothalamic contribution and/or further characterisation of this aspect is warranted.

3. qRT-PCR data presented in Figure 6A shows negligible alteration of HESX1 expression at all time points in NFKB2 mutant organoids. This is not consistent with the 2-fold increase in HESX1 expression described in day 48 organoids found by bulk RNA sequencing. How do the authors reconcile these results and why is one result focused on in the discussion where a potential mechanism for a blockade of normal pituitary cell differentiation is suggested? Further confirmation of HESX1 expression is required.

4. Throughout the authors focus on POMC gene expression and ACTH antibody immunopositive as being indicative of corticotroph cell identity. In the human fetal pituitary melanotrophs are present and most ACTH antibodies are unable to distinguish these cells from corticotrophs. Is the antibody used specifically for ACTH rather than other products of the POMC gene? It is unlikely that all the ACTH-positive cells are melanotrophs, nevertheless, it is important to know what the proportions of the 2 POMC-positive cell types are. This could be distinguished by looking for the expression of NeuroD1, which would also define whether corticotrophs are committed but not fully differentiated in the NFKB2 mutant organoids. In support of an effect on corticotrophs, it is notable that CRHR1 expression (which would be expected to be restricted to this cell type) is reduced by 84% in bulk RNAseq data (Table 1) and this may be an indicator of the loss of corticotrophs in the model.

5. Notwithstanding the caveats about whether the organoid model recapitulates *in vivo* pituitary differentiation (see 1 above) and whether the bulk RNAseq accurately reflects expression levels (see 3 above), there are potentially some extremely interesting changes in gene expression shown in Table 1 which warrant further discussion. For example, there is a 25-fold reduction in POU1F1 expression which may be expected to reflect a loss of somatotrophs in the organoid (and possibly lactotrophs) and highlights the importance of characterising the effect of NFKB2 on other anterior pituitary cell types within the organoid. If somatotrophs are affected, this may be relevant to the organoids as a model of DAVID syndrome as GH deficiency has been described in some individuals with NFKB2 mutations. The huge increase in CGA expression may reflect a switch in cell fate to gonadotrophs, as has been described with a loss of TPIT in the mouse. These are examples of the changes that warrant further characterisation and discussion.

6. How do the authors explain the lack of effect of NFKB2 mutation on global NFKB signalling?